Section 4.6 Graphs & geometric counting

图论与几何计数

[AIME-2002-I-Q05] | Part A (Q1-Q5)

Answer: 183

1. Each pair among twelve vertices determines three squares: two with the pair as a side and one with it as a diagonal. This gives $3\binom{12}{2} = 198$ incidences.

2. If three square vertices lie on the dodecagon circumcircle, the fourth lies on the same circle. Successive square vertices are $90^\circ = 3 \cdot 30^\circ$ apart, so the fourth is also a dodecagon vertex. There are exactly three such squares, obtained by taking every third vertex.

3. Each of those is counted six times, rather than once. Subtract $3(6-1)$ to get 183.

Source / 来源

Section 4.6 Graphs & geometric counting

图论与几何计数

[AIME-2004-I-Q03] | Part A (Q1-Q5)

Answer: 241

1. All vertex pairs number $\binom{26}{2} = 325$. Remove the 60 edges.

2. Triangular faces have no face diagonals; the twelve quadrilaterals have two each. Convexity prevents one such diagonal from belonging to two distinct faces. Thus remove another 24, leaving 241.

Source / 来源

Section 4.6 Graphs & geometric counting

图论与几何计数

[AIME-2007-II-Q05] | Part A (Q1-Q5)

Answer: 888

1. The intercept rectangle is 223×9, containing 2007 unit squares. Since $\gcd(223, 9) = 1$, the segment meets no interior lattice point, and it crosses $223 + 9 - 1 = 231$ square interiors.

2. The remaining 1776 squares split equally above and below by half-turn symmetry about the rectangle center. Thus the count is 888.

Source / 来源

Section 4.6 Graphs & geometric counting

图论与几何计数

[AIME-2014-I-Q05] | Part A (Q1-Q5)

Answer: 134

1. A separating circle intersects the common circumcircle in at most two points, unless the circles coincide; coincidence cannot strictly separate a proper nonempty subset. Thus any such subset is one consecutive run around the polygon.

2. Conversely any proper consecutive run can be strictly separated by a chord line placed in the two adjoining gaps. A sufficiently large circle approximating the appropriate half-plane separates the same finite vertex sets.

3. For each size one through eleven there are twelve runs. Add the empty and full subsets: $12 \cdot 11 + 2 = 134$.

Source / 来源

Section 4.6 Graphs & geometric counting

图论与几何计数

[AIME-2016-I-Q03] | Part A (Q1-Q5)

Answer: 810

1. A path enters the upper ring once, moves horizontally there, descends once to the lower ring, moves horizontally again, and then reaches B. It can never return to an earlier level.

2. On a five-cycle, a simple path from a fixed start has 1 zero-step option and two directions for each length $1, 2, 3, 4$, totaling 9. Changing horizontal direction would revisit a vertex.

3. There are 5 first upper vertices, 9 upper-ring paths, 2 descending edges, and 9 lower-ring paths. These choices are independent, giving $5 \cdot 9 \cdot 2 \cdot 9 = 810$.

Source / 来源

Section 4.6 Graphs & geometric counting

图论与几何计数

[AIME-2022-II-Q05] | Part A (Q1-Q5)

Answer: 072

1. For labels $i < j < k$, put $u = j - i, v = k - j$. All of $u, v, u + v$ must be prime. Parity forces exactly one of u, v to be 2, and the other must be p with $p, p + 2$ twin primes.

2. The possible p are $3, 5, 11, 17$. For each, the span is $p + 2$, allowing $20 - (p + 2)$ starting labels and two orders of the gaps.

3. Total $2(15 + 13 + 7 + 1) = 72$.

Source / 来源

Section 4.6 Graphs & geometric counting

图论与几何计数

[AIME-2025-II-Q03] | Part A (Q1-Q5)

Answer: 082

1. Choose the four central-cross colors first. Each small square has two internal edges and two independent outer edges; if its internal colors differ, its outer edges have two choices, otherwise only one.

2. For the central cross, monochromatic cases number 2, each weighted 1; cases with one odd-colored arm number 8, each weighted 4; two adjacent red arms number 4, each weighted 4; two opposite red arms number 2, each weighted 16.

3. Total $2+8\cdot 4+4\cdot 4+2\cdot 16=82$.

Source / 来源

Section 4.6 Graphs & geometric counting

图论与几何计数

[AIME-2008-II-Q10] | Part B (Q6-Q10)

Answer: 240

1. The nine possible squared distances are $1, 2, 4, 5, 8, 9, 10, 13, 18$, so at most ten points occur. A ten-point path must use these in that order.

2. Count backwards. The last two points are opposite corners, with four ordered choices. The preceding squared distance thirteen allows two symmetric choices. Fix the last points as $P_{10} = (0, 0), P_9 = (3, 3), P_8 = (0, 1)$. Distances ten,nine,eight,five,four,two successively force $P_7 = (3, 2), P_6 = (0, 2), P_5 = (2, 0), P_4 = (1, 2), P_3 = (1, 0), P_2 = (2, 1)$; the alternative ten-distance choice $(3, 0)$ fails at the next step because its distance-three neighbors were used.

3. The first point has three unused unit-neighbors of $(2, 1)$: $(1, 1), (2, 2), (3, 1)$. Thus ten-point paths exist and $r = 4 \cdot 2 \cdot 3 = 24$, giving $mr = 240$.

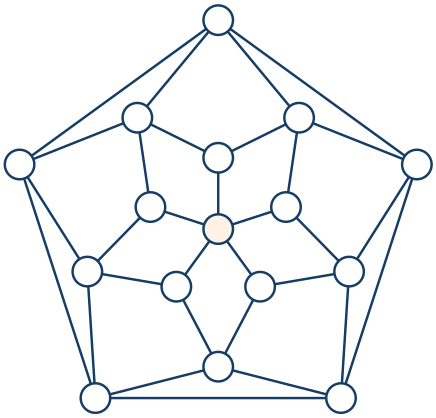
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Section 4.6 Graphs & geometric counting

图论与几何计数

[AIME-2012-I-Q07] | Part B (Q6-Q10)

Answer: 280



1. Let c_v be the coins at vertex v and d_v its degree. Fixed-state transfer gives $c_v = \sum_{u \sim v} c_u/d_u$. Thus c_v/d_v is the average of these same ratios at neighboring vertices.

2. A maximal ratio must equal all neighboring ratios; connectedness propagates this equality throughout the graph. Consequently c_v is proportional to d_v.

3. The central degree is five; the five inner degrees are three, and the ten other degrees are four. Total degree is $5 + 5 \cdot 3 + 10 \cdot 4 = 60$. The center receives $3360(5/60) = 280$ coins.

Source / 来源

Section 4.6 Graphs & geometric counting

图论与几何计数

[AIME-2018-I-Q07] | Part B (Q6-Q10)

Answer: 052

1. Each single base contributes six triangles with sides $1, 1, \sqrt{3}$ and two equilateral triangles, or eight. Both bases give sixteen.

2. For a mixed triangle choose which base contains the pair, in two ways. A cross-base distance is at least two, while base-pair lengths are $1, \sqrt{3}, 2$. Adjacent base pairs cannot be isosceles: no opposite-base vertex is horizontally equidistant from their endpoints.

3. For the six base pairs of length $\sqrt{3}$, two opposite-base vertices lie over their perpendicular bisector, giving twelve. For the three diameters of length two, a top vertex directly over either endpoint makes one cross-base side equal two, giving six; there are no equidistant top vertices for these diameters.

4. Total $16 + 2(12 + 6) = 52$.

Source / 来源

Section 4.6 Graphs & geometric counting

图论与几何计数

[AIME-2022-II-Q09] | Part B (Q6-Q10)

Answer: 244

1. Choosing two points on each line determines exactly one pair of crossing segments. Hence the interior crossing count is $X = \binom{7}{2}\binom{5}{2} = 210$.

2. Trim both full lines beyond the marked points; this does not alter bounded faces. For general counts m, n, the connected graph has $V = m + n + X + 4$ vertices and $E = (m + 1) + (n + 1) + mn + 2X$ edges.

3. Euler gives bounded faces $E - V + 1 = mn + X - 1$. Substituting $m = 7, n = 5$ gives $35 + 210 - 1 = 244$.

Source / 来源

Section 4.6 Graphs & geometric counting

图论与几何计数

[AIME-2012-II-Q14] | Part C (Q11-Q15)

Answer: 016

1. A simple graph with all degrees two is a disjoint union of cycles of length at least three. On nine vertices, the possible cycle sizes are $3+3+3, 3+6, 4+5, 9$.

2. An undirected cycle on k fixed labels has $(k-1)!/2$ arrangements. The four cases give $9!/[(3!)^3 3!] = 280$, $\binom{9}{3}(5!/2) = 5040$, $\binom{9}{4}(3!/2)(4!/2) = 4536$, and $8!/2 = 20160$.

3. The total is 30016, leaving remainder 16.

Source / 来源

Section 4.6 Graphs & geometric counting

图论与几何计数

[AIME-2017-II-Q11] | Part C (Q11-Q15)

Answer: 544

1. For five vertices, failure of strong connectivity forces an all-outgoing source or all-incoming sink. Indeed, a proper set of vertices reachable from a vertex has all cross edges entering it. A reachable set of size1 or2 contains a global sink; if its size is3 or4, the complementary set of size2 or1 contains a global source.

2. An all-outgoing vertex can be chosen5 ways, with the other6 edges free:320 orientations. There is at most one such vertex. Similarly there are320 with an all-incoming vertex.

3. The intersection has5 choices of source,4 of sink and3 free remaining edges, giving160. Out of $2^{10} = 1024$ orientations, the valid count is $1024 - 320 - 320 + 160 = 544$.

Source / 来源

Section 4.6 Graphs & geometric counting

图论与几何计数

[AIME-2017-II-Q13] | Part C (Q11-Q15)

Answer: 245

1. At each apex there are $\lfloor (n-1)/2 \rfloor$ symmetric vertex pairs. Every nonequilateral isosceles triangle has one apex, but an equilateral is counted three times. Thus $f(n) = n\lfloor (n-1)/2 \rfloor - (2n/3)\mathbf{1}_{3|n}$.

2. Write $n = 6k + j$ for $j = 0, 1, 2, 3, 4, 5$. Substituting into this formula gives successive differences $13k, 3k, 5k + 1, 7k + 3, 9k + 6, -k - 2$, respectively.

3. Equaling78 yields only $k = 6, j = 0$; $k = 26, j = 1$; and $k = 8, j = 4$. Therefore $n = 36, 157, 52$, with sum245.

Source / 来源

Section 4.6 Graphs & geometric counting

图论与几何计数

[AIME-2017-II-Q14] | Part C (Q11-Q15)

Answer: 168

1. Choose a primitive integral direction vector. Eight points span seven primitive steps, so every component has absolute value at most1. Axis-parallel lines contain10 points and do not qualify.

2. If exactly one component is zero, the line lies in one of30 coordinate-parallel 10×10 planes. In each plane, two diagonals of each slope $+1, -1$ have exactly8 points (offset magnitude2). This contributes120.

3. If all components are nonzero, there are four directions up to sign, equivalent by cube symmetry. For $(1, 1, 1)$, normalize a point on the line so its least coordinate is1. Exactly8 points means its greatest coordinate is3. The count of triples in $\{1, 2, 3\}^3$ using both1 and3 is $27 - 8 - 8 + 1 = 12$.

4. The space-diagonal contribution is $4 \cdot 12 = 48$, for total $120 + 48 = 168$.

Source / 来源

Section 4.6 Graphs & geometric counting

图论与几何计数

[AIME-2023-I-Q14] | Part C (Q11-Q15)

Answer: 608

1. Label states $(a, b) \in \mathbb{Z}_{12}^2$, and let $h(a, b)$ be 1 when the next move increases a, or 0 when it increases b. A Hamiltonian cycle must give every state one incoming edge. Thus $h(a-1, b) + [1 - h(a, b-1)] = 1$, so $h(a-1, b) = h(a, b-1)$.

2. Therefore h is constant on each diagonal $a + b \pmod{12}$. A valid transition rule is a binary word of length 12, read repeatedly as the diagonal sum increases one per move.

3. If the word contains c moves of the first hand, every twelve moves change the state by $(c, -c)$. Return occurs after $12/\gcd(c, 12)$ such blocks; all 144 states form one cycle exactly when $\gcd(c, 12) = 1$. This condition is sufficient as well as necessary.

4. The possible counts are $c = 1, 5, 7, 11$, giving $\binom{12}{1} + \binom{12}{5} + \binom{12}{7} + \binom{12}{11} = 1608$ sequences. The remainder is 608.

Source / 来源

Section 4.6 Graphs & geometric counting

图论与几何计数

[AIME-2024-II-Q15] | Part C (Q11-Q15)

Answer: 315

1. On the unit circumcircle, chord directions form twelve classes separated by 15°. There are six perpendicular direction pairs, three of each of two types. In a direction class of the first type, chord offsets are $0, \pm 1/2, \pm\sqrt{3}/2$; in the second they are $\pm\cos 75^\circ, \pm\cos 45^\circ, \pm\cos 15^\circ$.

2. Choose two chords in each perpendicular class. If D_1, D_2 are the maximum absolute offsets of each selected pair, their intersections lie on the chord segments exactly when $D_1^2 + D_2^2 \le 1$. For the five-chord class, selected pairs have $D = 1/2, \sqrt{3}/2$ with counts $3, 7$. The allowed ordered combinations number $3^2 + 2 \cdot 3 \cdot 7 = 51$.

3. For the six-chord class, the three possible D values have pair counts $1, 5, 9$. Every combination with the smallest D is allowed, as is the middle-middle combination. Thus the count is $1 + 2 \cdot 5 + 2 \cdot 9 + 25 = 54$. Total $3(51 + 54) = 315$.

Source / 来源

Section 4.6 Graphs & geometric counting

图论与几何计数

[AIME-2025-II-Q11] | Part C (Q11-Q15)

Answer: 113

1. Fix the common chord step $k = 1, \ldots, 12$. For $k < 12$, edges joining i to $i \pm k$ form $g = \gcd(24, k)$ cycles of length $24/g$. An even cycle has two perfect matchings and an odd cycle none.

2. For $k = 1, 5, 7, 11$ contributions are 2 each; $k = 2, 10$ give 4 each; $k = 3, 9$ give 8 each; $k = 4$ gives 16; $k = 6$ gives 64; $k = 8$ gives zero. The diameter case $k = 12$ contributes one.

3. Summing yields $8 + 8 + 16 + 16 + 64 + 1 = 113$. Different steps have different lengths, so no matching is duplicated.

Source / 来源

Section 4.6 Graphs & geometric counting

图论与几何计数

[AIME-2026-II-Q15] | Part C (Q11-Q15)

Answer: 393

1. Work over residues $0, 1, -1$. The triples $\{i, i+1, i+3\}$ are the seven Fano-plane lines. Let k be the number of zero positions; only lines avoiding those positions contribute to the cubic sum.

2. For $k = 7, 6, 5, 4, 3$, the respective counts are $1, 0, 42, 56, 42$: two nonzero entries must have opposite signs; three must be equal and not form a line; four must split two and two and the zero triple must itself be a line.

3. For $k = 2$, the five signs split four and one; the two surviving lines meet once, and their products cancel exactly when the minority sign is at one of four nonintersection positions. Count $\binom{7}{2} \cdot 2 \cdot 4 = 168$. For $k = 1$, the six positions correspond to edges of K_4 on the four surviving lines. Three negative signs must form a three-edge path to make exactly two line products negative; there are $4!/2 = 12$ paths per zero position, giving 84. With no zero entries the linear condition forces two or five negative signs, giving respectively four or three negative line products, never a zero cubic sum.

4. Total $1 + 42 + 56 + 42 + 168 + 84 = 393$. An independent exhaustive check can inspect all $3^7 = 2187$ residue tuples using the two displayed sums.

Source / 来源